

# Reproducible ML Project Notes

## Purpose

This sample document summarizes basic practices for reproducible machine learning projects. The practices are relevant for small research prototypes as well as larger applied AI projects.

## Reproducibility checklist

Experiments should use Git version control, a requirements.txt file, a configuration file, and a fixed random seed when randomness is used. The train/validation/test split should be saved or documented. Evaluation metrics should be written to a file, and important outputs should be reproducible from the command line.

## Testing and code organization

Small tests are useful for functions such as chunking, PDF loading, metric calculation, feature extraction, and data cleaning. The aim is not to test whether a deep model always gets the same final accuracy, but to catch silent errors in preprocessing and evaluation code.

## LLM and RAG notes

For document question answering, a compact RAG pipeline can extract text from PDFs, split text into chunks, create embeddings, store vectors in FAISS, retrieve relevant chunks, and send the retrieved context to an LLM. The final answer should be grounded in the retrieved context and should cite source chunks when possible.